

Jason Ie

949-849-2950 | ie.jasonray10@gmail.com | [linkedin.com/in/jason-ie/](https://www.linkedin.com/in/jason-ie/) | github.com/jason-ie | jasonie.com

EDUCATION

Northeastern University | Boston, MA

May 2026

Bachelor of Science in Computer Science and Finance, Minor in Mathematics

GPA: 3.85/4.0

- **Relevant Coursework:** Algorithms and Data Structures, Object Oriented Programming, Software Engineering, Networks and Distributed Systems, Database Design, Artificial Intelligence, Algorithmic and Robo Trading, Mathematics of Data Models, Discrete Mathematics, Blockchain Applications
- **Leadership & Activities:** Co-founder of Huskies for Hoops (nonprofit), Mentor for Computer Science Mentoring Organization, HackPrinceton, Quant-Finance Club, Disrupt Fintech Initiative
- **Honors:** Dean's Scholarship, Dean's List

SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, C++, SQL, GraphQL, C#, Kotlin, HTML, CSS

Frameworks: Spring Boot, React, Next.js, Node.js, Express.js, FastAPI, Django, PyTorch, Pandas, Scikit-learn, Jest

Cloud & Databases: AWS, GCP, PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch, Supabase, Firebase

Tools: Git, Docker, Kubernetes, Jenkins, LocalStack, Linux, TCP/IP, Postman, Selenium, Jira, Figma, Render, Claude Code

EXPERIENCE

Associate Software Engineer

Aug 2026 - Present

Capital One

Chicago, IL

- Engineering a modernized storage system for sensitive credit card data using **Java**, **Go**, **PostgreSQL**, and **AWS**, upgrading legacy card security infrastructure for Card Tech's Card Management Security team
- Developing **PIN update** and **PIN mailing** services behind **API Gateway** for Capital One cardholders
- Implementing secure **PIN generation** and **encryption**, protecting PINs in transit and at rest during data migration

Undergraduate Quantitative Research Assistant

Feb 2026 - May 2026

Northeastern University, D'Amore-McKim

Boston, MA

- Designed an algorithmic signal generation framework in **Python** and **Pandas** to exploit time-series and cross-sectional momentum, analyzing return auto-covariance and lead-lag effects across historical market data
- Constructed an automated feature-engineering and model-training pipeline with **Scikit-learn** and **PyTorch** to train **gradient boosting algorithms** for momentum signal generation, dynamically modeling for frictions like market impact

Software Engineer Intern

Jun 2025 - Aug 2025

Capital One

McLean, VA

- Architected a backend for data usage and metadata governance using **Java Spring Boot** and **AWS DynamoDB**, accelerating executive-level compliance processes across enterprise operations to save **\$2M+** annually
- Automated **CI/CD pipelines** with **Docker** and **Jenkins** for containerized **LocalStack** environments, streamlining **Agile** workflows and reducing development setup time by **60%**
- Engineered a responsive frontend architecture with **React** and **TypeScript**, integrating **TanStack Query** for asynchronous state management and data caching to eliminate redundant network requests

Software Developer Co-op

Jul 2024 - Dec 2024

enLabel Global Services

Boston, MA

- Led the end-to-end development of a client-facing **Next.js** and **React** registration portal to onboard **10,000+** users, implementing an **Express.js** middleware layer with **OAuth 2.0 (PKCE)** for secure Google Drive and Dropbox integrations
- Optimized **SQL** queries and refactored stored procedures in **SSMS**, reducing query execution time across a **1,000+** table database serving **50,000+** active users

Software Engineer Intern

Dec 2023 - Apr 2024

SuperCare Health

Irvine, CA

- Optimized a **data analysis** pipeline by introducing **webhook-based** event processing, reducing latency by **25%** for critical patient data workflows
- Modernized the legacy fax system by integrating the **RingCentral API** with **MongoDB**, capable of managing **20,000+** faxed pages daily across patient communication channels

PROJECTS

Vera | Python, FastAPI, PostgreSQL/pgvector, Gemini Vision

Feb 2026

- Orchestrated a multi-stage **computer vision** pipeline to automate fashion discovery, reducing manual user search time to seconds by chaining **Gemini Vision** with a **Segment Anything Model (SAM)** for precise object extraction
- Reduced product search latency by over **85%** by engineering a vector-based matching engine with an asynchronous **FastAPI** backend, **PostgreSQL/pgvector**, and scalable **CLIP image embedding** search